

Sahayak · सहायक: prototype submission (Ideas for India 2026)

An offline scam shield and benefits guide for people who are new to digital money. One small computer at a CSC or bank-agent counter runs everything. Phones join its Wi-Fi, which has no internet, and ask in Hindi or English, by voice or touch: *"Is this a scam?"* and *"What am I owed?"* Nothing a person types or says leaves the node.

Team: Roshan Raj and Ishmit Singh · Problem statement: Inclusive Innovation for Bharat

What is in this submission

File	What it is
01_Sahayak_Demo_Video.mp4	2 min 12 s: the problem, the five main flows on screen, the measured results
02_Sahayak_Pitch_Deck.pdf / .pptx	12 slides in the challenge's outline
03_Sahayak_OnePager.pdf	Everything on one A4 page
04_Application_to_Prototype.pdf	What changed since the application, and why each change was made
05_Sahayak_Testing_Report.pdf	How every number was measured, with confidence intervals and limits
06_Sahayak_Jury_Kit.pdf	Printable test cards (messages, UPI QR codes, benefit personas) with an answer key
07_Sahayak_Source_Code.zip	The full source: node, phone app, console, content packs, benchmarks, 360 tests
screenshots/ , clips/	14 phone screenshots and 5 silent screen recordings

See it in 2 minutes

Watch 01_Sahayak_Demo_Video.mp4 . The clips show the real app running on the node: a fake KYC SMS, a benefits interview for a 67-year-old widow, a "scan to get cashback" UPI QR, the operator console, and the node page counting Sahayak's internet connections (zero).

Run it yourself (about 15 minutes, once)

Needs Python 3.10 or newer on Windows, macOS or Linux. No GPU. About 250 MB of speech models are downloaded once (300 MB more for screenshot reading). On Linux, install the CPU-only PyTorch first, as requirements-ocr.txt explains.

```
unzip 07_Sahayak_Source_Code.zip && cd sahayak
python -m pip install -r requirements.txt
python -m pip install -r requirements-ocr.txt # optional: reading screenshots (adds PyTorch)
python scripts/get_models.py                # once, with internet: speech and screenshot-reading models
python -m sahayak                           # the node; prints its address and the console PIN
```

To try it on a phone with no node at all, open <https://roshworldwide.github.io/optum-v2/> (the stand-alone build, python scripts/build_tryit.py); after the first visit it works with the internet off.

Open `http://<node-address>:8000` on a phone on the same Wi-Fi (or `http://127.0.0.1:8000` on the same computer). The operator console is `/app/console.html` and the node's status page is `/app/node.html`. After the models are downloaded, the internet can be switched off: everything keeps working. `python -m pytest` runs the 360 tests.

Try it in 5 minutes

On your phone: open <https://roshworldwide.github.io/optum-v2/>, check a message, then switch on airplane mode and check another. It still works: the scam check and the benefits interview run on the phone itself.

Print `06_Sahayak_Jury_Kit.pdf`. Type, speak or photograph a message card; photograph a QR card in "QR code" mode; answer the benefits interview as one of the persona cards. Try to fool it with your own message. The last page shows what Sahayak answered when the kit was built. The demo QR codes use a UPI handle that does not exist, so a real UPI app refuses them.

Measured, not claimed

Area	Result
Scam detection (frozen test split, 60 messages)	91.9% of scams caught vs 51.4% for a keyword blocklist; false alarms 8.7% vs 43.5%
Red team (52 scams disguised to get past it)	46 caught on the first run vs 33 for the blocklist; 52 after the fixes it found
Scheme rules (200 personas × 12 schemes)	2,400 / 2,400 decisions match an independent re-derivation; median 5 questions
Hindi speech (Google FLEURS, 418 recordings)	15.4% word error, offline on a laptop CPU
Speed	2 ms per verdict; 0.65 s from releasing the mic to the spoken reply
Offline	0 connections from Sahayak to the internet, counted live by an audit hook

Our test messages were written by our team, so these numbers are optimistic. The testing report says what each number does not show. Next: real messages collected with consent, voice recordings from older speakers, and a field test at a CSC with an operator.

Privacy and safety

Messages, voices and answers stay in the node's memory for at most 15 minutes and are never written to disk. The case log needs consent, holds no names or numbers, is encrypted and deletes itself after 30 days. "No signs found" never says "safe". Every amount shown comes from an Ed25519-signed content pack; an altered pack is refused.

Code: MIT licence. The Hindi voices and the optional language model are licensed for non-commercial use only.